The feasible region satisfying all the conditions is OABCD.

The co-ordinates of the points are O(0,0); A(20,0); B(20,10); C(18,12) and D(0,12).

| Corner points | $Z = 2x_1 + 3x_2$ |
|---|---|
| O(0,0) | 0 |
| A(20,0) | 40 |
| B(20,10) | 70 |
| C(18,12) | 72 |
| D(0,12) | 36 |

Table 10.4

Maximum value of $Z$ occurs at C. Therefore the solution is $x_1 = 18$ , $x_2 = 12$, $Z_{max} = 72$.

**Example 10.8**

Maximize $Z = 3x_1 + 4x_2$ subject to $x_1 - x_2 \leq -1$; $-x_1 + x_2 \leq 0$ and $x_1, x_2 \geq 0$.

***Solution:***

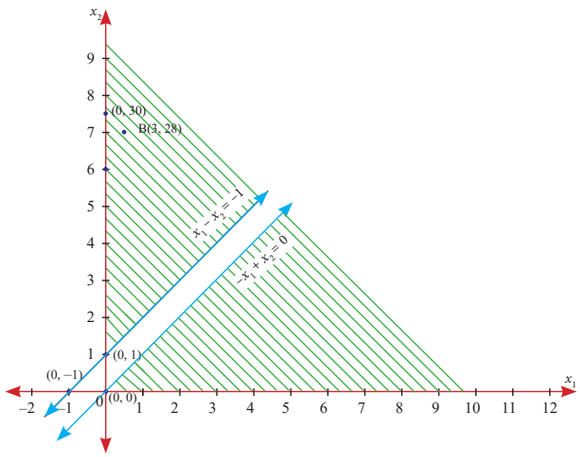



Fig 10.4

Since both the decision variables $x_1$, $x_2$ are non-negative ,the solution lies in the first quadrant of the plane.

Consider the equations $x_1 - x_2 = -1$ and $-x_1 + x_2 = 0$.

$x_1 - x_2 = -1$ is a line passing through the points (0,1) and (–1,0).

$-x_1 + x_2 = 0$ is a line passing through the point (0,0).

Now we draw the graph satisfying the conditions $x_1 - x_2 \leq -1$; $-x_1 + x_2 \leq 0$ and $x_1, x_2 \geq 0$.

There is no common region(feasible region) satisfying all the given conditions. Hence the given LPP has no solution.

## Exercise 10.1

1. A company produces two types of pens A and B. Pen A is of superior quality and pen B is of lower quality . Profits on pens A and B are ₹ 5 and ₹ 3 per pen respectively. Raw materials required for each pen A is twice as that of pen B. The supply of raw material is sufficient only for 1000 pens per day . Pen A requires a special clip and only 400 such clips are available per day. For pen B, only 700 clips are available per day . Formulate this problem as a linear programming problem.

2. A company produces two types of products say type A and B. Profits on the two types of product are ₹ 30/- and ₹ 40/- per kg respectively. The data on resources required and availability of resources are given below.

| | Requirements | | Capacity available per month |
|---|---|---|---|
| | Product A | Product B | |
| Raw material (kgs) | 60 | 120 | 12000 |
| Machining hours / piece | 8 | 5 | 600 |
| Assembling (man hours) | 3 | 4 | 500 |

Formulate this problem as a linear programming problem to maximize the profit.

3. A company manufactures two models of voltage stabilizers viz., ordinary and auto-



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